

SIMATS ENGINEERING

ASSESSMENT TOOL I – Uncertainty Visualization Lab

Course: Data Handling and Visualization	Code: DSA06	CO: C05
Duration: 180 min	Total Marks: 100	Reg No : 192424238 Name: Jayaprakash Nivas

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

I Jayaprakash Nivas (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Understanding of Uncertainty 25%	Visualization of Uncertainty 25%	Application of Visualization Principles 20%	Organization 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Bootstrapping and Confidence Intervals – Point Estimate

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data: 31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: resampling, point estimate, confidence interval, histogram/density plot, interpretation.

Code :

```
data<-c(31,35,28,42,37,33,45,29,39,34,41,36)

set.seed(123)

n<-5000

boot_means<-replicate(n,mean(sample(data,replace=TRUE)))

mean_value<-mean(data)

ci<-quantile(boot_means,c(0.025,0.975))

print(paste("Sample Mean:",round(mean_value,2)))

print(paste("95% Bootstrap CI:",round(ci[1],2),"to",round(ci[2],2)))

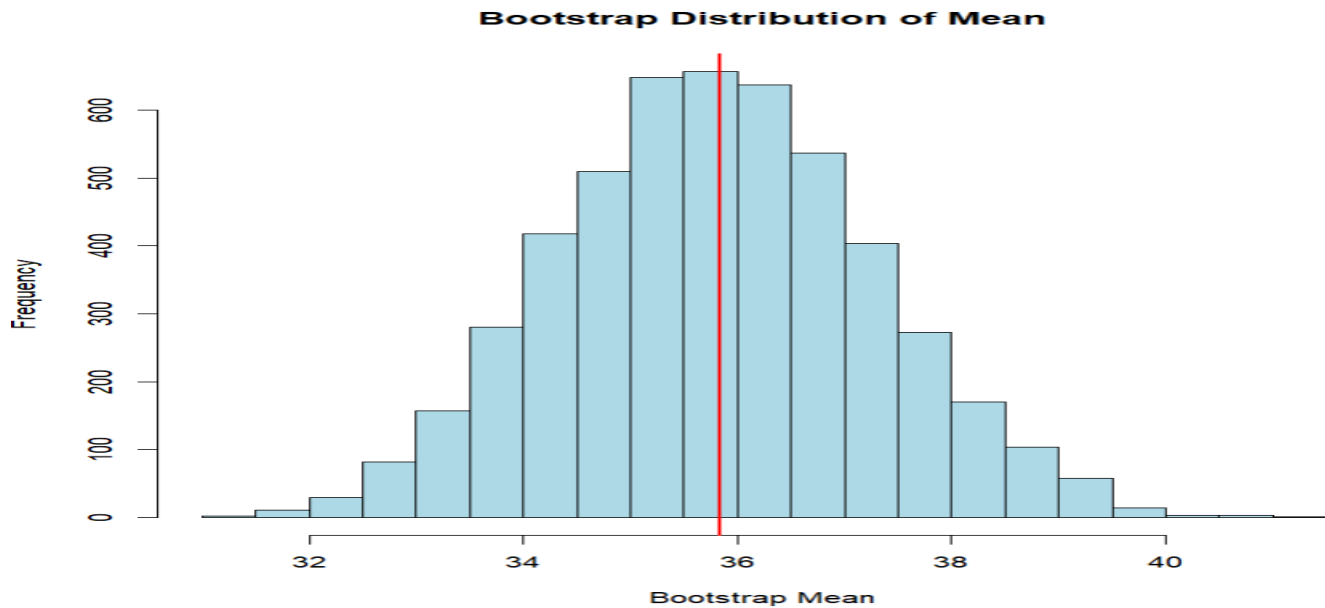
hist(boot_means,breaks=30,main="Bootstrap Distribution of Mean",xlab="Bootstrap Mean",col="lightblue")

abline(v=mean_value,col="red",lwd=2)
```

Output :

"Sample Mean: 35.83"

"95% Bootstrap CI: 33.08 to 38.75"



Interpretation

The sample mean delivery time is approximately 35.83 minutes. The 95% bootstrap confidence interval gives a range of plausible values for the population mean delivery time. The bootstrap distribution shows the uncertainty associated with estimating the mean from the sample.

Q2. Uncertainty in Trend Estimation

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band.

Explain how the uncertainty changes across the range of advertising expenditure.

Data: Advertising = 10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40;

Sales = 48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

Expected R tasks: lm(), prediction interval/confidence interval, ggplot2 visualization, trend interpretation.

Code :

```
library(ggplot2)

advertising<-c(10,12,15,18,20,23,25,28,30,33,36,40)

sales<-c(48,51,53,57,60,61,66,69,72,75,79,84)

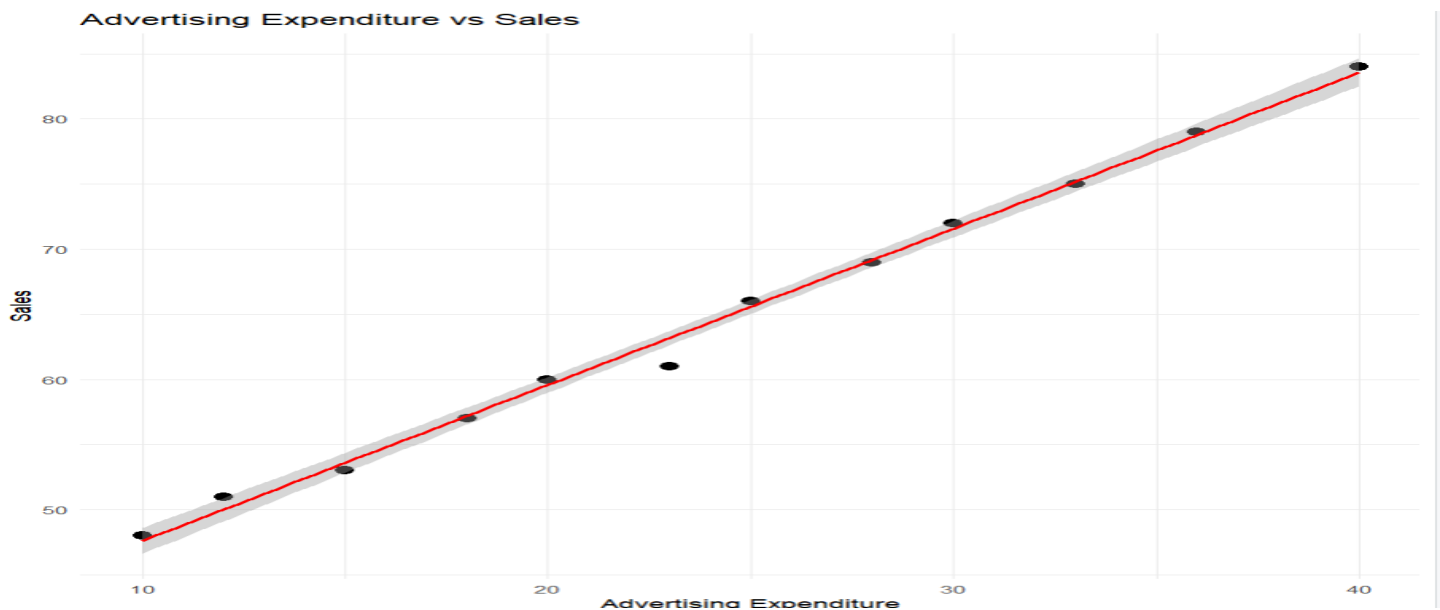
data<-data.frame(advertising,sales)

model<-lm(sales~advertising,data=data)

print(summary(model))

ggplot(data,aes(x=advertising,y=sales))+geom_point(size=3)+geom_smooth(method="lm",se=TRUE,color="
red")+labs(title="Advertising Expenditure vs Sales",x="Advertising
Expenditure",y="Sales")+theme_minimal()
```

Output :



Interpretation

There is a strong positive relationship between advertising expenditure and sales. Sales generally increase as advertising expenditure increases. The 95% confidence band represents uncertainty around the estimated regression trend and is generally narrower near the center of the observed advertising range and wider toward the extremes.

Q3. Overplotting, Jittering and Transparency

A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation.

Data: Hours = 2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,8,8,8,8,8,9,9,9,10,10;

Scores = 45,52,52,58,60,60,64,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94

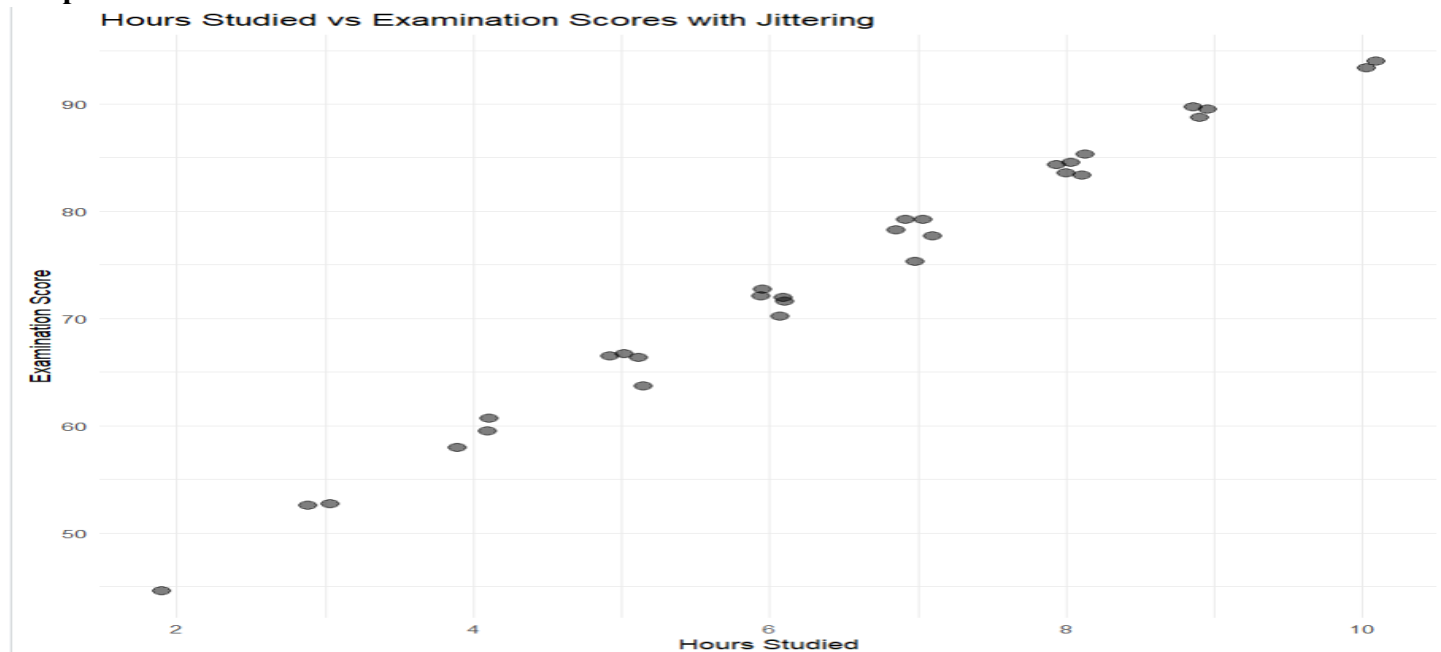
Expected R tasks: `geom_point()`, `alpha`, `position_jitter()`, comparison of overplotting.

Code :

```
library(ggplot2)
data<-
data.frame(hours=c(2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,8,8,8,8,8,9,9,9,10,10),scores=c(45,52,52,58,60,60,64,
66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94))
ggplot(data,aes(x=data$hours,y=data$scores))+geom_point(size=3)+labs(title="Hours Studied vs Examination
```

```
Scores",x="Hours Studied",y="Examination Score")+theme_minimal()
ggplot(data,aes(x=data$hours,y=data$scores))+geom_jitter(width=0.15,height=0.8,alpha=0.5,size=3)+labs(title
="Hours Studied vs Examination Scores with Jittering",x="Hours Studied",y="Examination
Score")+theme_minimal()
```

Output :



Interpretation

The original scatterplot suffers from overplotting because several students have identical values. Jittering separates overlapping observations slightly, while transparency helps show areas with greater concentration. Therefore, the redesigned plot provides a clearer representation of the data.

Q4. 2D Histogram and Density Estimation

A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

Data: Height = 150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186;

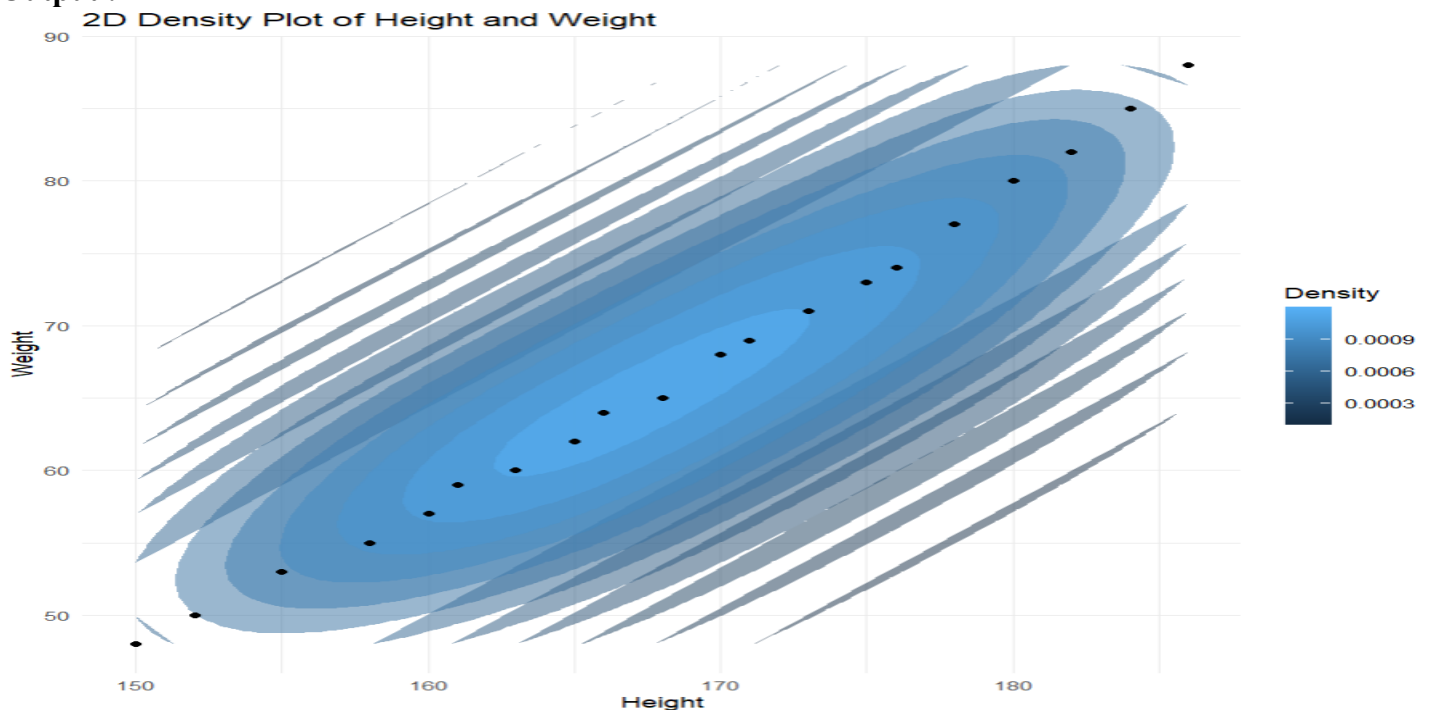
Weight = 48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88

Expected R tasks: 2D binning, density estimation, contour/density layers, interpretation of concentration.

Code :

```
library(ggplot2)
data<-
data.frame(height=c(150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186),weight=c(48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88))
ggplot(data,aes(x=height,y=weight))+geom_bin2d(bins=6)+labs(title="2D Histogram of Height and Weight",x="Height",y="Weight",fill="Count")+theme_minimal()
ggplot(data,aes(x=height,y=weight))+stat_density_2d(aes(fill=after_stat(level)),geom="polygon",alpha=0.5)+geom_point()+labs(title="2D Density Plot of Height and Weight",x="Height",y="Weight",fill="Density")+theme_minimal()
```

Output :



Interpretation

The highest observation density is concentrated around the middle range of height and weight, where several observations occur close together. The 2D density plot communicates concentration more clearly because the density contours show the areas where observations are most concentrated, while the 2D histogram divides the data into fixed bins.

Q5. Color Scales, Accessibility and Misleading Visualization

The following category-wise satisfaction scores must be visualized in R. Create two plots: (1) a plot using a nonmonotonic/diverging color scheme that could mislead interpretation, and (2) a redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale. Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data: Category = A,B,C,D,E,F,G;

Satisfaction = 42,48,55,61,67,74,82

Expected R tasks: color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.

Suggested R packages: ggplot2, dplyr, boot (or base R).

Assessment focus: Correct R code, appropriate uncertainty visualization, interpretation, and identification of misleading design choices.

Code :

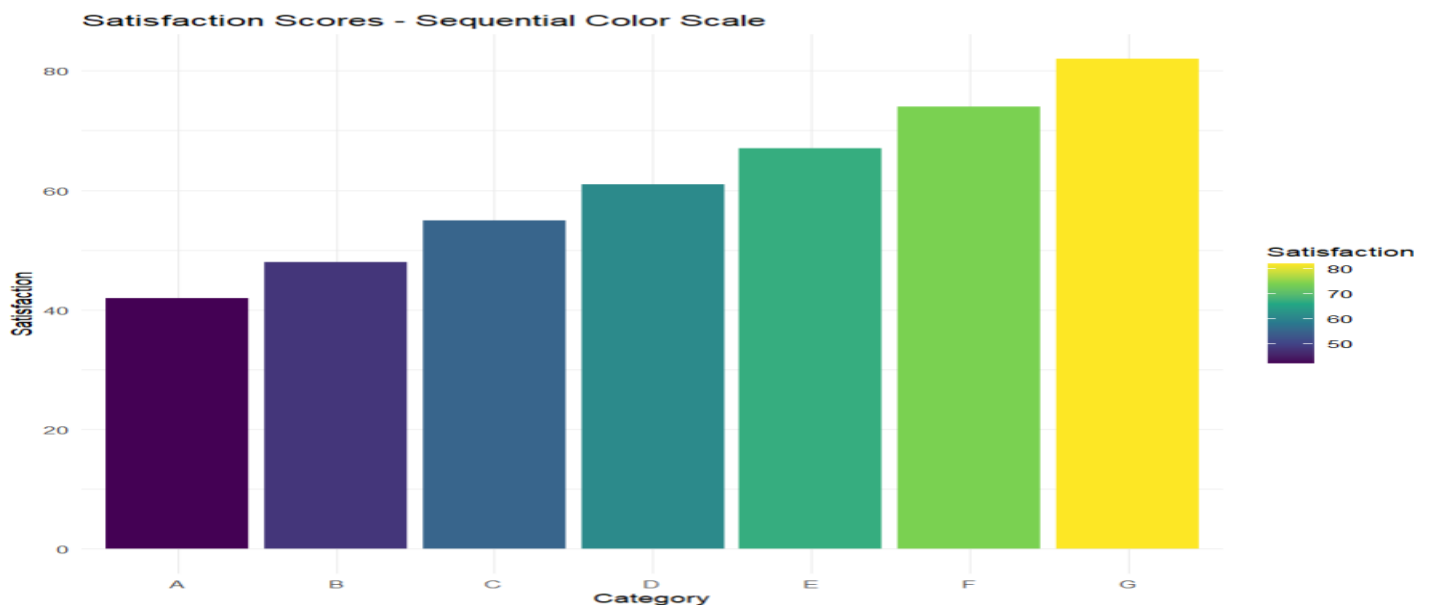
```
library(ggplot2)
```

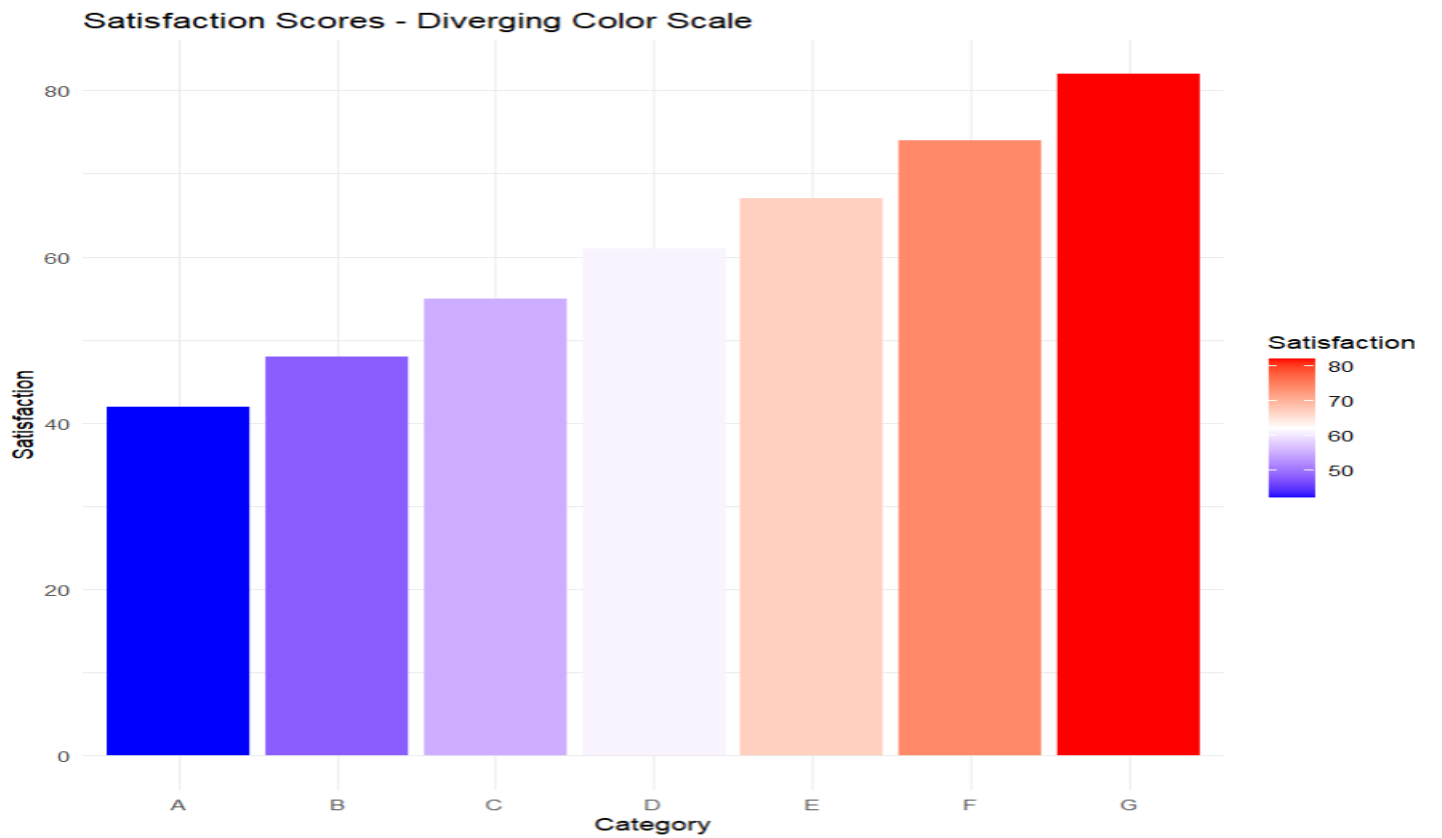
```
data<-data.frame(category=c("A","B","C","D","E","F","G"),satisfaction=c(42,48,55,61,67,74,82))
```

```
ggplot(data,aes(x=category,y=satisfaction,fill=satisfaction))+geom_col()+scale_fill_gradient2(low="blue",mid="white",high="red",midpoint=62)+labs(title="Satisfaction Scores - Diverging Color Scale",x="Category",y="Satisfaction",fill="Satisfaction")+theme_minimal()
```

```
ggplot(data,aes(x=category,y=satisfaction,fill=satisfaction))+geom_col()+scale_fill_viridis_c(option="D")+labs(title="Satisfaction Scores - Sequential Color Scale",x="Category",y="Satisfaction",fill="Satisfaction")+theme_minimal()
```

Output :





Interpretation

The first plot uses a diverging color scale, which introduces contrasting colors and can imply an artificial midpoint or meaningful separation in the satisfaction values. This can mislead the viewer because satisfaction is an ordered numerical variable.

The second plot uses an ordered sequential scale, where the color intensity changes consistently with satisfaction. It is easier to interpret and is more suitable for viewers with color-vision deficiencies. Therefore, the second design communicates the increasing satisfaction values more accurately.